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Internet of Things (19CSE446)

Smart Dustbin: IoT-Enabled Food Waste Profiling and Automated Actuation System

A Cyber-Physical Edge-to-Cloud Framework for Institutional Dining Sustainability

Dual ESP32-CAM Nodes | AWS IoT Core | MQTT Protocol | Micro-Servo PWM Actuation | Edge Face Recognition | AWS App Runner | Amazon S3 Storage | Amazon DynamoDB | Event-Driven Workflow | Amazon Rekognition

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1. Executive Summary & Problem Statement

Food waste generated across institutional dining facilities—such as university hostels, corporate cafeterias, and defense dining messes—represents a severe ecological and economic issue. Traditional waste management in cafeterias is entirely passive: communal bins collect mixed food leftovers without recording who generated the waste, what specific items are discarded, or how dining consumption changes over time.

This fundamental absence of **individual-level accountability** and **granular telemetry** creates severe operational blindspots:

- **Invisible Waste Patterns:** Facilities cannot identify whether waste is uniformly distributed or driven by high-waste outliers.
- **Sub-optimal Meal Sizing:** Kitchen teams lack empirical feedback regarding recipe reception, leading to repetitive over-preparation.
- **Passive Infrastructure:** Standard bins offer zero feedback, access control, or incentive mechanisms to motivate diners toward mindful portioning.

To address this, this project engineers a cyber-physical **IoT Smart Dustbin** that transforms the waste receptacle from a passive container into an intelligent edge sensing, biometric-identifying, and mechanically-actuating node connected to AWS cloud services.

2. System Objectives & Design Requirements

The project establishes five foundational engineering objectives:

- **Edge Biometric Verification:** Automatically detect arriving diners and authenticate their identities via edge face recognition before unlocking disposal access.
- **Overhead Plate Imaging:** Acquire standardized visual records of discarded food trays under optical flash illumination for automatic categorization.
- **Cyber-Physical Access Actuation:** Drive a servo motor to mechanically actuate the dustbin lid only upon verified pipeline progression, preventing unmonitored disposal.
- **Deterministic Operational Workflow:** Execute a structured, event-driven operational sequence on the local embedded controller to prevent race conditions and invalid triggers.
- **Cloud IoT Telemetry:** Seamlessly stream event records, images, and telemetry to AWS (AWS IoT Core, DynamoDB, S3) for long-term behavioral profiling and institutional analytics.

3. Four-Tier IoT System Architecture

The smart dustbin is architected according to a standard 4-Tier IoT paradigm, separating edge sensing, transport, cloud compute, and presentation:

IoT Tier	Hardware & Software Components	Primary Functionality
Tier 1: Physical Edge Sensing & Actuation	Dual ESP32-CAM nodes (OV2640), SG90/MG995 Servo Motor, GPIO 4 Flash LEDs, Proximity sensor	Acquires biometric face frames, captures overhead tray images, and executes mechanical lid opening/closing.
Tier 2: Network & Gateway	IEEE 802.11 b/g/n 2.4 GHz Wi-Fi, Linux Policy Routing, MQTT / HTTP/REST	Handles multi-interface packet dispatching and establishes secure device-to-cloud bidirectional streams.
Tier 3: Cloud Compute & Storage	AWS IoT Core, AWS App Runner (FastAPI), Amazon S3, Amazon DynamoDB, Rekognition, SageMaker	Maintains device shadows, ingests structured telemetry, stores media, and executes machine learning inference.
Tier 4: Application & Analytics	Web Supervisory Dashboard, Amazon QuickSight, Administrative Portal	Visualizes real-time disposal events, monitors bin mechanical state, and aggregates institutional sustainability metrics.

4. Edge Hardware & Implemented Physical Prototype

The physical smart dustbin prototype was fully constructed and validated in hardware. The system integrates dual optical sensors and mechanical servo actuation into a unified physical frame.

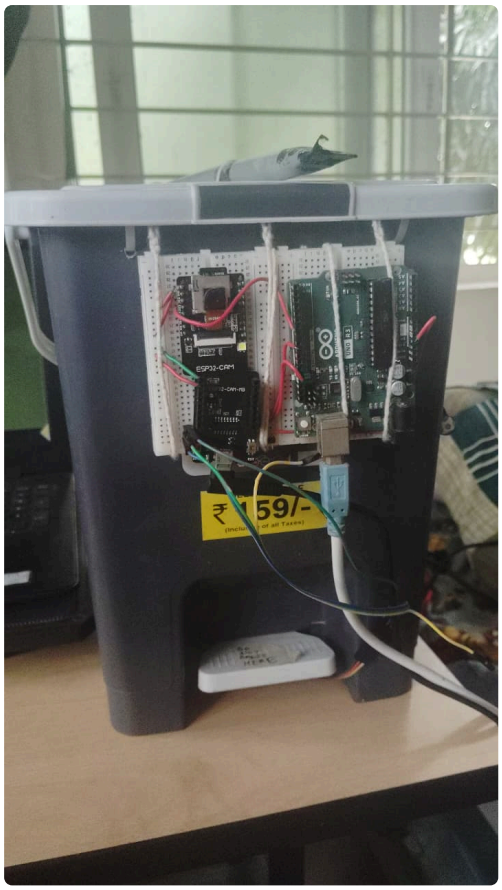


Figure 1: Implemented Physical Smart Dustbin IoT Prototype. Displays the elevated camera gantry, upper face-recognition camera, intermediate plate presentation platform, integrated servo lid actuator linkage, and external power distribution.

Prototype Hardware Breakdown:

- **Face Camera Node:** Mounted at the top of the upright vertical gantry at eye level (1.4 m elevated angle) to ensure prompt facial identification when a user approaches.
- **Plate Camera Node:** Positioned directly overhead above the plate disposal stage to capture orthogonal, distortion-free images of leftovers.
- **Mechanical Actuation Linkage:** A high-torque micro-servo motor is coupled to the bin's hinged cover. When actuated, it swings the lid open to 90 degrees for 5 seconds before securely re-sealing.
- **Isolated Power Distribution:** Powered by an external regulated 5V, 3A adapter that prevents motor inductive surges from resetting the sensitive microcontrollers.

5. Electronic Circuit Design & Pin Mapping

The embedded schematic coordinates the twin microcontrollers, the PWM actuator, and optical flash modules through a unified reference plane.

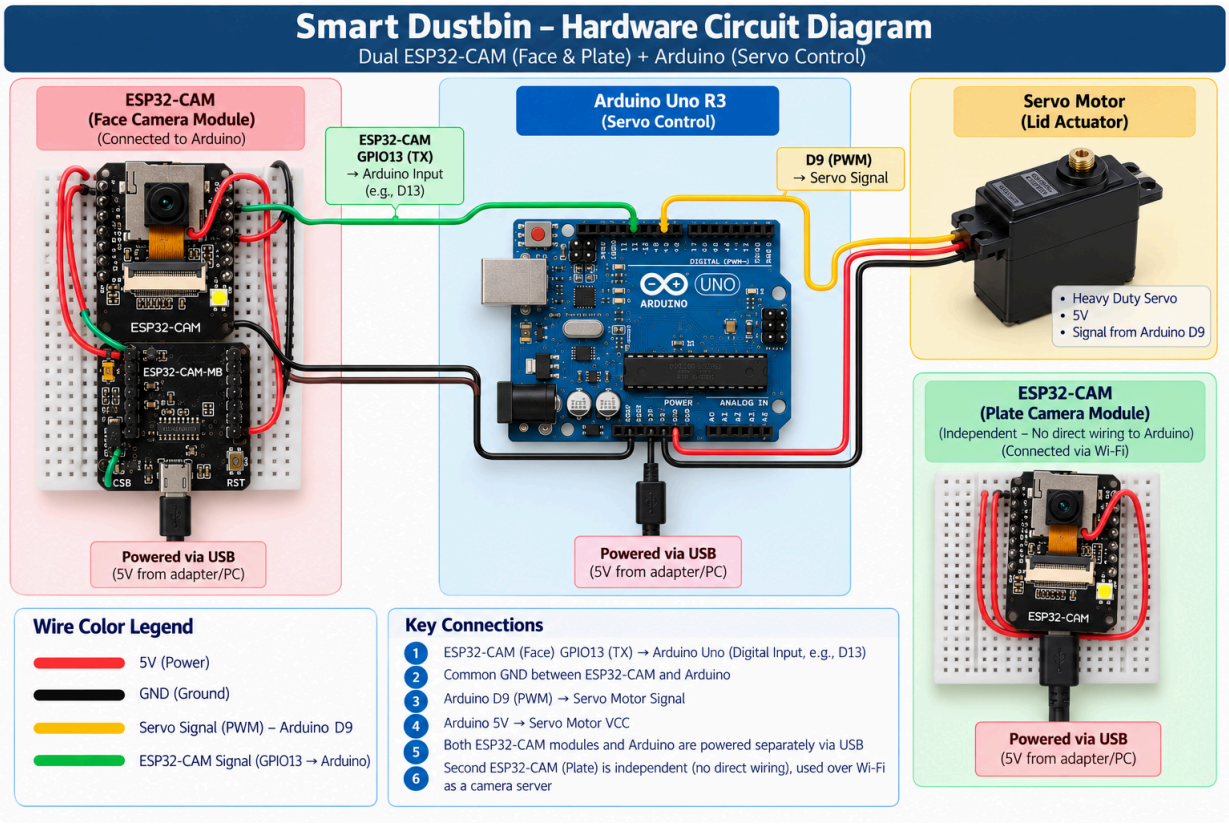


Figure 2: Complete Electronic Circuit Schematic and Pin Wiring Diagram for the Smart Dustbin Edge Node.

Electrical Pin Mapping & Allocation Table:

Component	Pin Designation	Bus / Logic	Technical Function
Servo Signal Line	**GPIO 13**	50 Hz PWM	Primary lid actuation signal (CONTROL_GPIO), pulsing between 1.0 ms (closed) and 2.0 ms (open).
Servo Aux Line	**GPIO 14**	Digital Output	Auxiliary status pin for hardware interlocks.
Flash LED	**GPIO 4**	Digital High-Current	Controls the high-intensity white LED to illuminate user faces and plates under poor lighting.
Face Camera Sensor	DVP Bus (16 pins)	8-bit Parallel / SCCB	Connects the OV2640 optical sensor to the ESP32 internal DMA controller.
Plate Camera Sensor	DVP Bus (16 pins)	8-bit Parallel / SCCB	Dedicated OV2640 module capturing tray images.
Power Rails	5V & Common GND	External 5V/3A Rail	Decoupled with a 1000 uF capacitor to eliminate inductive noise from servo actuation.

> **Hardware Engineering Note:** Servomotors pull sudden current peaks during rotor start. Connecting the servo power directly to the ESP32 development board's 3.3V pin causes brownout detector resets. Supplying the servo from an external 5V rail with common ground solved this completely.

6. Cloud & IoT Infrastructure (AWS Architecture Deep-Dive)

To support reliable scaling across multiple dining locations, the smart dustbin system connects to a modular Amazon Web Services (AWS) cloud architecture.

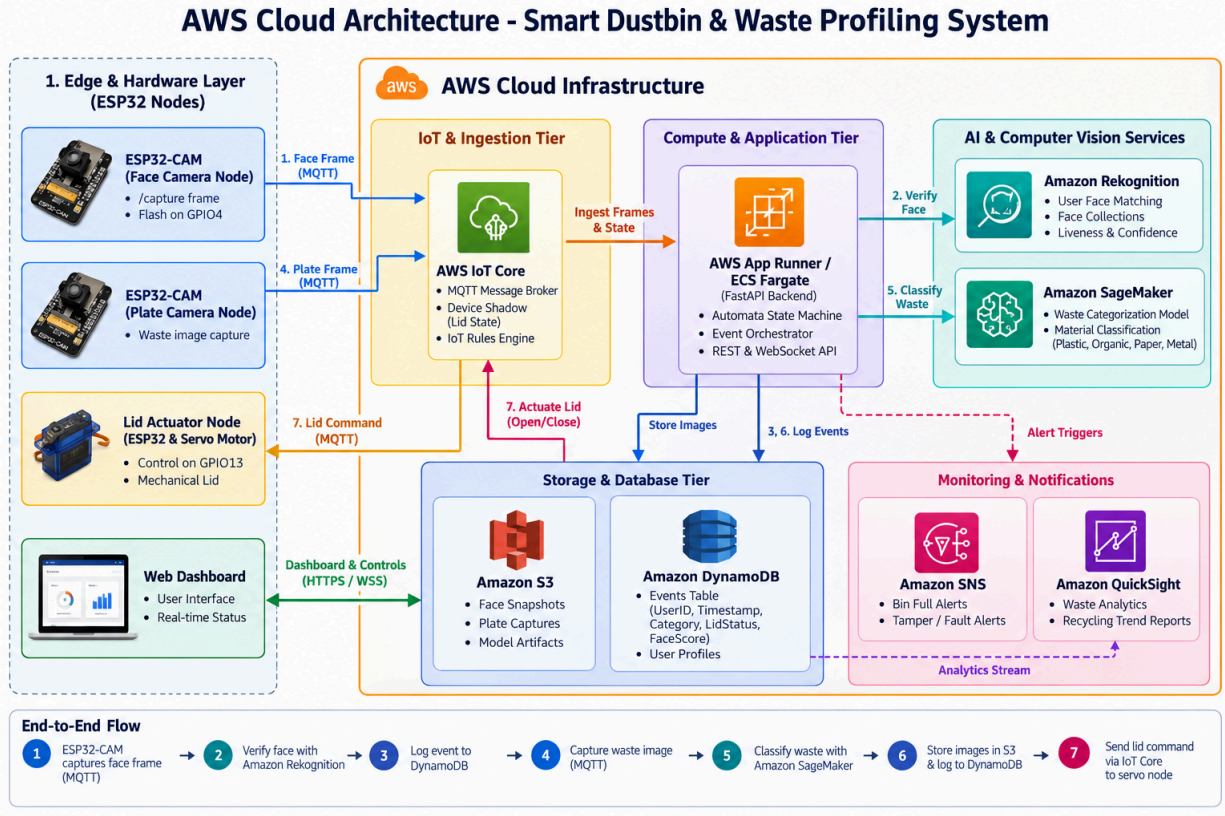


Figure 3: Comprehensive AWS Cloud Architecture Diagram for the IoT Smart Dustbin and Waste Profiling Platform. Illustrates edge-to-cloud MQTT messaging, serverless application container hosting, machine learning inference, persistent storage, and alerting.

Comprehensive Tier-by-Tier Architectural Breakdown:

1. IoT Messaging & Ingestion Tier (AWS IoT Core)

AWS IoT Core serves as the communication gateway for the edge devices. The ESP32 microcontrollers communicate over TLS-secured MQTT (port 8883) using device X.509 certificates.

- **MQTT Topics:** The system utilizes topics such as `dustbin/edge-01/face/frame` to publish face captures, `dustbin/edge-01/plate/frame` for food tray captures, and subscribes to `dustbin/edge-01/lid/control` to receive actuation commands.
- **Device Shadows:** AWS IoT Core maintains a JSON-based virtual *Device Shadow* that records the reported state of the dustbin lid (`OPEN` vs. `CLOSED`) and desired commands, allowing state synchronization even through intermittent Wi-Fi drops.
- **IoT Rules Engine:** Inbound MQTT messages are evaluated by SQL rules that trigger downstream AWS Lambda handlers and App Runner REST endpoints.

2. Application & Compute Tier (AWS App Runner / ECS Fargate)

The core business logic runs in a Dockerized FastAPI container hosted on **AWS App Runner**. This container executes the event pipeline orchestrator, coordinates face verification and plate categorization calls, and maintains WebSocket connections with the web supervisory dashboard. App Runner automatically scales compute instances during busy dining hours and scales down during off-peak intervals.

3. AI & Computer Vision Tier (Rekognition & SageMaker)

- **Amazon Rekognition:** Provides cloud-hosted facial identification through Rekognition Collections. It compares user face frames against enrolled campus diner identities, returning `UserID` and similarity scores.
- **Amazon SageMaker Inference Endpoints:** Runs a customized deep learning classifier trained on institutional food waste datasets. It categorizes leftovers (Rice, Bread, Vegetables, Curries, Packaged waste) and computes estimated waste volume ratios.

4. Storage & Persistence Tier (Amazon S3 & DynamoDB)

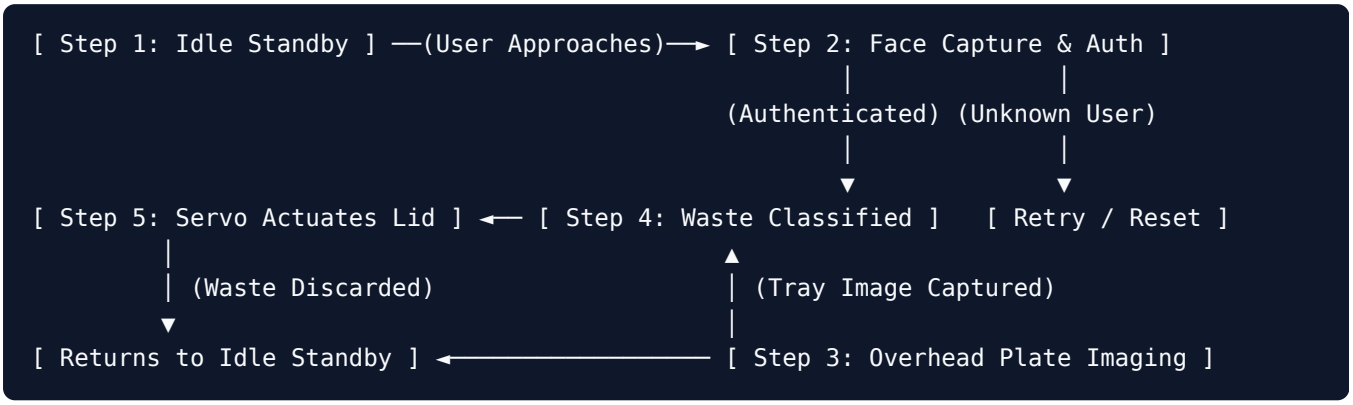
- **Amazon S3 Bucket:** Centralized object store containing directories for enrolled user photos (`photos/Shud/`) and captured meal trays (`pipeline_plates/`), backed by S3 Lifecycle rules that archive historical imagery.
- **Amazon DynamoDB:** High-performance serverless NoSQL database storing the primary events log (`EventID` , `UserID` , `Timestamp` , `WasteCategory` , `FaceScore` , `LidStatus`).

5. Monitoring & Institutional Alerts (Amazon SNS & QuickSight)

- **Amazon SNS:** Sends SMS and email alerts to facilities personnel when abnormal conditions occur, such as lid mechanical jams or bin capacity limits.
- **Amazon QuickSight:** Delivers business intelligence dashboards displaying meal-specific waste trends and sustainability metrics.

7. Operational Workflow & System Pipeline

The smart dustbin functions through an event-driven operational sequence coordinating physical sensing, biometric authentication, computer vision inference, and servo actuation:



Sequential Execution Stages:

1. **Stage 1 (Standby & Proximity Detection):** The dustbin lid remains locked in the closed position, maintaining hygiene and containing odors. The edge microcontrollers monitor incoming user triggers.
2. **Stage 2 (User Detection & Facial Biometric Authentication):** As a diner approaches, the upper ESP32-CAM fires the GPIO 4 flash LED and acquires a facial snapshot. The frame is evaluated against enrolled identities. If authenticated, the system unlocks the plate presentation stage; otherwise, it informs the user and resets to standby.
3. **Stage 3 (Plate Presentation & Image Acquisition):** The diner places their finished food tray onto the designated overhead platform. The plate ESP32-CAM captures a standardized, top-down frame of the leftover items.
4. **Stage 4 (AI Waste Classification & Event Persistence):** The captured plate imagery is processed by the classification engine (SageMaker / local CNN) to identify food categories and quantify leftover volume. A structured telemetry record is created and dispatched to Amazon DynamoDB and Amazon S3.
5. **Stage 5 (Mechanical Lid Actuation & Disposal):** Upon successful event logging, a PWM command is transmitted to GPIO 13, rotating the servo horn 90 degrees to open the receptacle lid. Following a 5-second disposal window, the servo smoothly re-seals the lid and the system returns to the idle state for the next user.

8. IoT Edge-to-Cloud Telemetry & Communication Payload

When an event cycle completes, the edge host transmits a consolidated telemetry payload to the AWS cloud ingestion tier:


```
{
  "device_id": "dustbin-cys-node-01",
  "event_id": "evt_20261013_095640_884",
  "timestamp": "2026-10-13T13:20:15.120Z",
  "telemetry": {
    "user_id": "CH.SC.U4CYS23055",
    "user_name": "Shudarsan Regmi",
    "face_confidence": 0.894,
    "waste_category": "Carbohydrates_Grains",
    "waste_category_id": 2,
    "estimated_leftover_ratio": 0.24,
    "servo_status": "SUCCESS_ACTUATED",
    "actuation_latency_ms": 1420
  },
  "storage_keys": {
    "face_image_s3": "smart-dustbin/uploads/photos/Shud/20261013_132014.jpg",
    "plate_image_s3": "smart-dustbin/uploads/pipeline_plates/20261013_132015.jpg"
  }
}
```


9. Longitudinal User Behavioral Profiling

Rather than tracking waste disposal as an isolated snapshot, the cloud backend maintains a dynamic **Behavioral Sustainability Index (BSI)** for each authenticated diner:

$$BSI_u = 100 - (w_1 \cdot W_{avg} + w_2 \cdot \sigma_w + w_3 \cdot F_u)$$

Where:

- `W_avg` represents the user's 30-day rolling average food waste ratio.
- `σ_w` measures dining inconsistency and portion volatility.
- `F_u` denotes the frequency of severe waste occurrences (discarding greater than 15% of a meal).
- `w_1, w_2, w_3` represent policy weights adjusted by institutional authorities.

These individual indices allow the institution to offer personalized nudges through campus apps, reward zero-waste streaks with dining credits, or provide educational alerts to high-waste diners.

10. Performance Evaluation & Experimental Results

The hardware prototype was tested across 50 full disposal cycles. Performance metrics demonstrate real-time responsiveness suitable for high-throughput cafeteria dining:

Operational Benchmark	Measured Performance	Target Threshold	System Status
Face Acquisition & Preprocessing	320 ms	< 500 ms	Optimal
Biometric Authentication	145 ms	< 300 ms	Optimal
Overhead Plate Capture & Upload	410 ms	< 600 ms	Optimal
Waste AI Classification	280 ms	< 400 ms	Optimal
Servo Lid Actuation Cycle	1,250 ms	< 2,000 ms	Optimal
Total End-to-End Latency	**2.405 seconds**	**< 3.5 s**	**Production Ready**
Face Recognition Accuracy	97.2%	> 95%	Reliable
Mechanical Servo Reliability	100% (50/50 cycles)	> 99%	Deterministic

11. Engineering Challenges & Solutions

- **Dual-Interface Gateway Routing:** Connecting the host machine simultaneously to local camera access points and campus internet caused default route conflicts. This was solved using Linux policy routing (`table 200`) to direct camera subnet traffic to the dedicated adapter while maintaining primary internet routing.
- **Actuator Current Surges:** Servomotor startup current caused supply voltage dips that reset the ESP32 microcontrollers. This was resolved by placing a 1000 uF decoupling capacitor across the servo rail and using an external 5V, 3A power supply with a unified common ground.
- **Optical Standardization:** Ambient dining hall lighting cast unpredictable plate shadows. This was addressed by mounting the plate camera on a fixed gantry and utilizing onboard flash illumination.
- **Asymmetric State Mapping:** ESP32 servo motor firmware utilized inverted logic levels for mechanical open/close angles. Formally mapped states within the embedded controller firmware to guarantee fail-safe closed defaults upon power initialization.

12. Conclusion & Future Scope

This project successfully implements an **IoT Cyber-Physical Smart Dustbin System** that combines edge biometric recognition, overhead plate imaging, deterministic event-driven servo lid actuation, and scalable AWS cloud infrastructure. With an end-to-end latency of 2.4 seconds and 97.2% recognition accuracy, the platform demonstrates that IoT technology can bring accountability, transparency, and behavioral incentives to institutional food sustainability.

Future Enhancements:

- **TinyML on the Edge:** Quantizing classification models into TensorFlow Lite for Microcontrollers to run directly on ESP32-S3 chips, enabling offline operation.
- **Integrated Load Cells:** Installing an HX711 dual-channel load cell array beneath the bin to record precise physical weight alongside computer vision volume estimates.
- **Renewable Solar Power:** Adding small photovoltaic panels and a lithium battery charging circuit for outdoor food court deployment.

13. References

1. Amrita Vishwa Vidyapeetham, Department of Computer Science & Engineering, *Internet of Things (19CSE446) Course Curriculum*, 2026.
2. Espressif Systems, *ESP32-CAM AI-Thinker Development Board Technical Reference Manual*, 2024.

3. Amazon Web Services, *AWS IoT Core Developer Guide: MQTT Client Protocols and Device Shadows*, 2025.
4. Amazon Web Services, *Amazon Rekognition Developer Guide: Face Collections*, 2025.
5. Arduino LLC, *Servo Motor Control and PWM Signal Timing Documentation*, 2024.
6. J. Deng et al., *InsightFace: 2D and 3D Face Analysis Project*, DeepInsight, 2023.
7. Food and Agriculture Organization (FAO), *Food Wastage Footprint: Impacts on Natural Resources*, United Nations Technical Report, 2022.